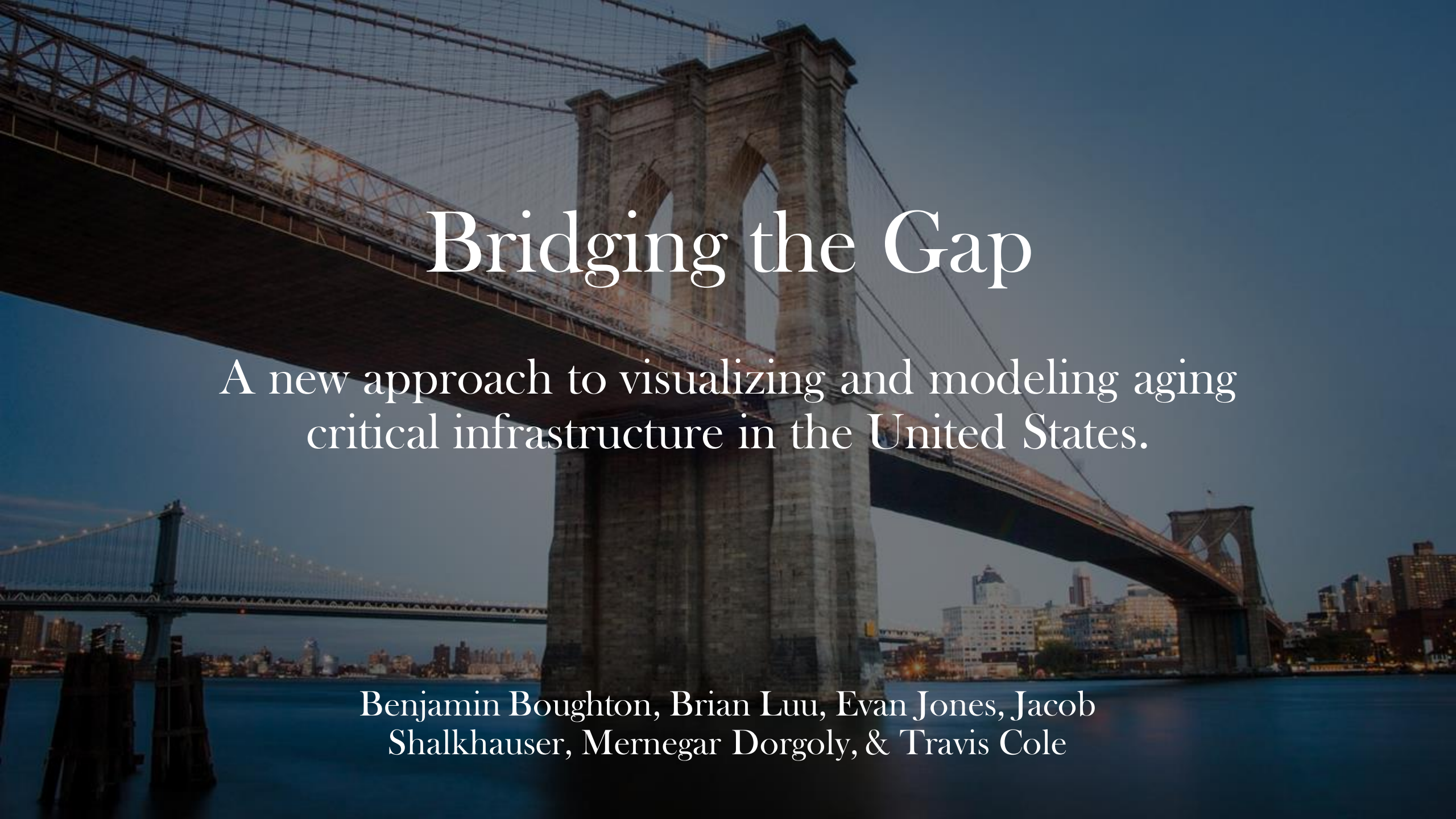




Bridging the Gap

A new approach to visualizing and modeling aging critical infrastructure in the United States.

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United States Bridge Infrastructure Today

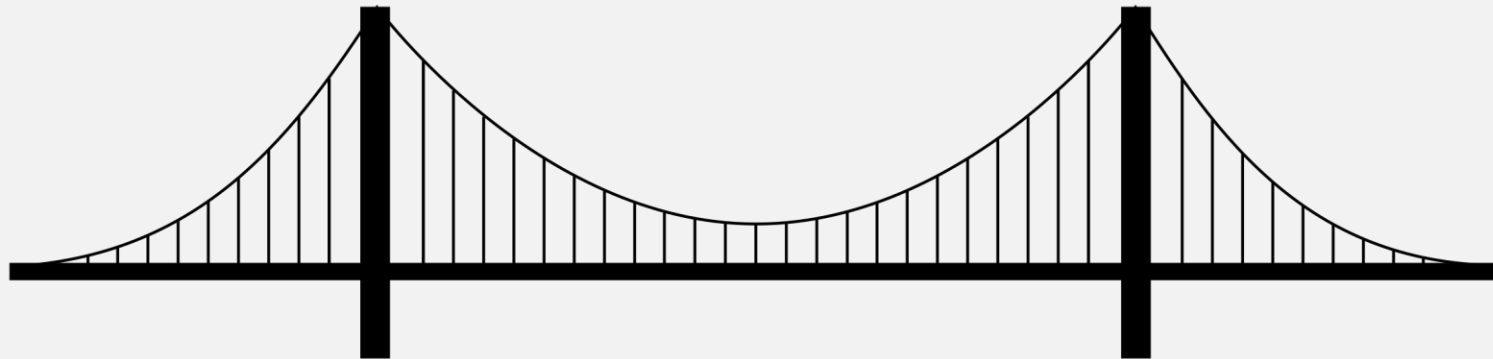
There are over 617,000 bridges in the United States.

- Each bridge is critical to the transportation of goods, services, and individuals.

Unfortunately, a significant number of these bridges are decades old and in a general state of disrepair^[1].

- Nearly half of all bridges were built more than 50 years ago.
- 55% of bridges are classified as being in "Fair" or "Poor" condition.

There has not been considerable action or discourse addressing deteriorating bridge structure due to the lack of open-source and transparent solutions for exploring the infrastructure data.



[1] NBI/FHA, "National bridge inventory dataset," <https://infobridge.fhwa.dot.gov/Data/Dashboard>, NBI/FHA, 2021, accessed: 2021-10-11.

The Current State and Limitations of Today

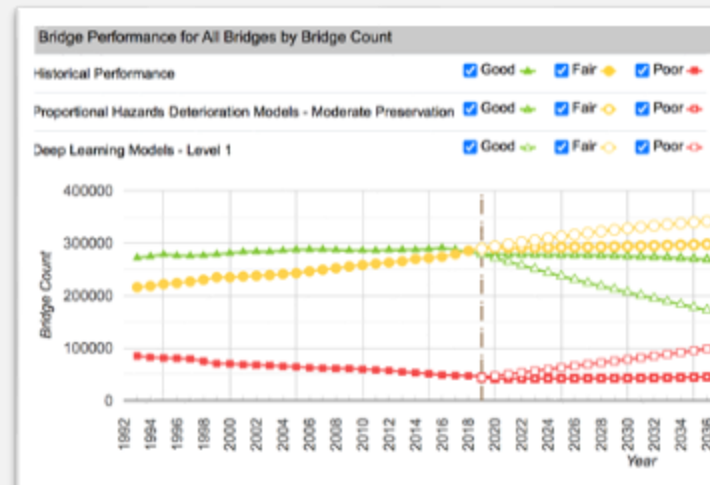
Leading tools for understanding and visualizing bridge data, such as "InfoBridge"^[23] from the U.S. Department of Transportation (DOT), are severely lacking in many aspects.

- Data is presented in a format that does not meet modern visualization design principles.
- Column names are confusing, making data non-accessible.
- The geographic mapping lacks interactivity and has no purpose beyond general location.
- There is limited potential to explore data due to non-user-friendly interface design decisions.
- Performance and projection charts are not visually appealing or easy to understand.

Selected Bridges				
1 - State Name	8 - Structure Number	22 - Owner Agency	3 - County Name	27 - Year Built
~	x	~	x	=
1 - Alabama	000677	State Highway Agency	065 - Hale County	1929
1 - Alabama	000676	State Highway Agency	065 - Hale County	1929
1 - Alabama	000675	State Highway Agency	043 - Cullman County	1929
1 - Alabama	000674	State Highway Agency	043 - Cullman County	1929
1 - Alabama	000673	State Highway Agency	065 - Hale County	1929
1 - Alabama	000671	State Highway Agency	129 - Washington Count	1929
1 - Alabama	000344	State Highway Agency	105 - Perry County	1926

Tabular data is difficult to filter and lacks visual clarity.

Performance and project charts are hard to read.



Geographic mapping is not interactive or very useful.

Our Objectives; The Improvements We Plan to Make

Our primary objective is to create a new framework for visualizing and modeling bridge inventory information in the United States; we will improve upon present solutions to create a product that includes **more accurate and comprehensive modeling approaches** as well as **more accessible and visually-appealing data**.

We will leverage the Google's Street View API, Python Flask, and various visualization libraries to create visually-appealing and interactive visualizations that improve upon the present solutions.

- Users will have a more localized and meaningful geographic visualization; drilling-down will lead to additional data about each bridge.

We will improve upon contemporary models by unifying various datasets (traffic_[22], climate_[21], & accident_[22]) that are serious contributors to the deterioration of bridge infrastructure.

What's New in Our Approach; Why We'll Succeed

Our approach primarily improves upon the existing solutions in two major categories: **data/visualization** and **modeling**.

Data/Visualization

- We will base our designs upon the core principles of data visualization.
- We will develop interactive visualizations including a choropleth detailing bridge health, interactive tool-tips, and Google Street View images of bridges ^[24].
- We will make use of interactive visualization tools (e.g. Bokeh, JavaScript, etc.)
- We will clean the data to eliminate known inconsistencies and issues.

Modeling

- We will include real-world ambient temperature^[18], climate^[19], and traffic/accident data^[20] in our models to create more consistent and accurate predictions.
- We aim to use an iterative design approach to prototype and test several models improving upon the existing classes of models currently used in the field.

Those Who Care; Who Our Application Will Impact

Our application will impact all individuals interested in clearly understanding the current state of bridge infrastructure in the U.S.

The American Road & Transportation Builders Association (ARTBA) states that there are over 230,000 bridges that need replacing or serious repairs^[26].

- Americans drive over these bridges 1.5 billion times per day.
- There are over 175 million daily crossings on structurally deficient bridges.

We believe our application will help U.S. travelers more easily understand the condition of the bridges they use every day; this will allow them to make better-informed decisions surrounding bridge usage and better hold their municipalities accountable regarding bridge maintenance, regulations, and repairs.

[26] J. Schneidawind, "230,000 U.S. bridges need repair, new analysis of federal data finds." The American Road Transportation Builders Association (ARTBA), p. 1, 2019. [Online]. Available: <https://www.artba.org/2020/04/12/230000-u-s-bridges-need-repair-new-analysis-of-federal-data-finds>

The Impact of Our Success and How We Measure It

Success of our application will result in individuals across the United States having a better fundamental understanding of the condition of bridges in their area.

- This could lead to increased infrastructure funding and more infrastructure-related political discourse; as this is often public data, we could easily measure these changes^[27].

Comparative user studies can be used to measure our visualization improvements, user-interface upgrades, Tufte's metrics^[25], and the enhancements to data access.

To measure the success of our models, we will utilize cross-validation to perform model selection and test accuracy on out-of-sample data.

The Risks and Payoffs

The most realistic risk faced by our project is the possibility of creating a model that is inaccurate and poorly performing.

If the public relies on an inaccurate model, they could be led to make misinformed and potentially dangerous decisions.

- Careful model design and cross-validation will reduce these risks.

If our model performs well and is highly accurate, the public will be able to depend on our data for any travel-related decision making in their daily lives.

The Cost and Timeline of Our Project

The data for this project is publicly-available, so the major costs will be associated with computation, infrastructure, and hosting.

- Amazon AWS



The project will take roughly 7 weeks to complete with the final deadline of the completed application being December 3rd of 2021.

How We Will Measure Our Progress

We will measure our progress via "midterm" and "final exam" deliverable checkpoints.

The "midterm" checkpoint will occur at roughly 3 weeks into project development.

- We will discuss isolated features of the project, talk with each of our three project teams (data, modeling, and visualization), and conduct integration of each team's work.

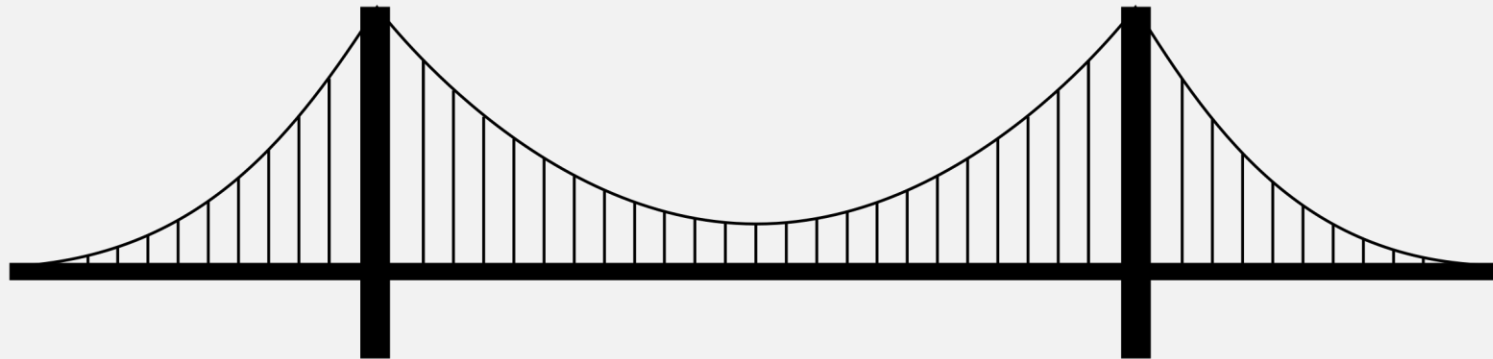
The "final exam" checkpoint will occur at roughly 6 weeks into project development.

- We will test the complete project integration and finish work on the project's final report and presentation.

Conclusion

Our team believes we can create a more accessible, more accurate, and more visually appealing alternative to today's existing U.S. bridge infrastructure dashboards.

By tackling this project with better visualizations and stronger models, the general public will be better informed on the current state of the bridges they use every day.



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